



Simulation of Gravity Data Needed for GPS-Denied Navigation

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Introduction

Global Navigation Satellite Systems (GNSS), including GPS, are used as a primary source of position for many navigation systems. The main limitation is that GNSS depends on radio frequency signals that can become unavailable or unreliable because of obstruction, interference, jamming, or spoofing. In these situations an Inertial Navigation System (INS) can continue to estimate position without receiving an external signal, but the INS solution accumulates error with time. This makes an independent method of correcting INS drift important for long periods of GNSS denial.

Gravity Aided Navigation uses the spatial variation of the Earth's gravity field as a reference for position. A gravimeter can measure gravity while a stored gravity map provides the expected gravity at known locations. If a measured gravity sequence can be matched to a similar sequence in the reference map, the resulting location estimate can be used to correct the INS. Gravity is attractive for this purpose because the signal is naturally available and does not require the vehicle to transmit an external signal. Previous studies have shown the feasibility of gravity aided navigation using correlation matching, Kalman filtering, ICCP, particle filtering, and other matching approaches [2]-[11].

The purpose of this research was to build a complete simulation that could test gravity map matching and then extend the problem to gravity-aware route planning. The first part of the work models an aircraft travelling across the Patterson Lake airborne gravity survey area and compares an INS only path with gravity corrected paths. The second part tests OMSL-ICCP as a sequence-based gravity matching method. The final part investigates whether A* path planning can be modified to select routes that pass through gravity regions expected to be more useful for navigation.

Background and Literature Review

Gravity Aided Navigation

Gravity aided navigation has commonly been investigated as a way to support inertial navigation when other positioning sources are limited. Wang et al. performed a South China Sea trial using measured marine gravity data, stored gravity maps, a multi-model Kalman filter, and GPS as reference truth. Their work showed that gravity information could improve an inertial solution and also showed that gravity map quality and gravity feature variation strongly affect positioning performance [6]. Xiong et al. investigated a cubature Kalman filter approach in which the difference between measured gravity anomaly and map-indicated gravity anomaly was used as a measurement for estimating navigation errors [7]. These studies support the use of gravity as an independent navigation reference but also show that performance depends strongly on the local gravity field.

Correlation based approaches such as TERCOM and ICCP solve the problem differently. Instead of directly estimating the full navigation state through a filter, they compare a measured gravity pattern with patterns stored in a map. This makes spatial distinctiveness important. If several locations have similar gravity values or similar contour shapes, a matching algorithm can return a low mathematical matching cost while still selecting the wrong geographic position.

OMSL-ICCP

Liu et al. proposed Gravity Aided Positioning Based on Real-Time ICCP With Optimized Matching Sequence Length [10]. Their method addresses the fact that the best ICCP matching sequence length can change with the gravity characteristics along a route. The method uses Golden Section Search to choose a sequence length, a Hausdorff-distance based search region, and a Gravitation Field Algorithm (GFA) for final refinement.

The paper reported that changing the sequence length can improve the performance of real-time ICCP when the gravity field changes along a trajectory [10].

The OMSL-ICCP code used in this project is a paper-based implementation of the method described by Liu et al. It is not a reproduction of the authors' original source code. Several implementation details that were not fully specified in the paper, including the exact contour extraction procedure and some stopping details, were developed for this simulation. This distinction is important when comparing the results in this report with the published results.

Gravity-Aware Route Planning

Wang and Cai proposed a path planning method for underwater gravity aided inertial navigation using the Posterior Cramer-Rao Bound (PCRB) and A* [12]. Their method includes both gravity information and predicted INS uncertainty when evaluating routes. The study showed that a route selected for navigation accuracy can be slightly longer than a shortest route while improving the expected navigation performance [12]. This provided the basis for investigating a similar idea in this project.

The exact combination of the A* variations developed in this project with OMSL-ICCP is a proposed adaptation. It should not be described as a published A* and OMSL-ICCP algorithm. The work uses the general gravity-aware path-planning idea from Wang and Cai [12] and the OMSL-ICCP matching method from Liu et al. [10], then tests whether route selection can improve sequence-based gravity matching.

Data

Patterson Lake Airborne Gravity Data

The simulation uses airborne gravity data from the Patterson Lake survey area. The datasets were obtained from the Natural Resources Canada Geoscience Data Repository [1]. Two numerical gravity grids were used. The primary layer is the Patterson Lake Vertical Free Air Gravity Anomaly. The second layer is the First Vertical Derivative of the Free Air Gravity Anomaly.

Table 1: Patterson Lake gravity dataset properties

Property	Free Air Gravity Anomaly	First Vertical Derivative
Raster size	344 x 294 cells	344 x 294 cells
Grid spacing	500 m	500 m
Left / Right	492,750 m / 664,750 m	492,750 m / 664,750 m
Bottom / Top	6,348,250 m / 6,495,250 m	6,348,250 m / 6,495,250 m
Minimum	-55.91 mGal	-44.55
Maximum	5.38 mGal	63.85
Mean	-23.79 mGal	0.094
Standard deviation	11.97 mGal	11.23
NoData	-999999 converted to NaN	Aligned with Free Air grid

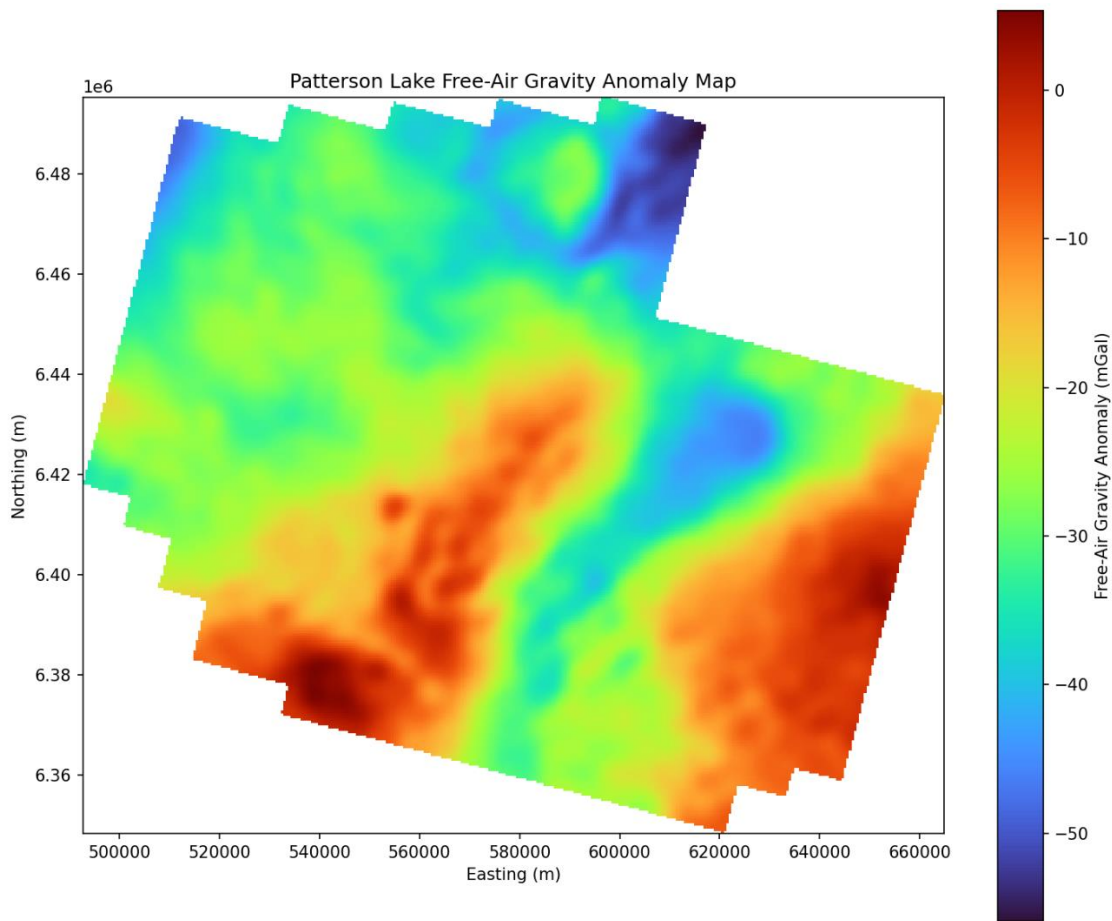


Figure 1: Patterson Lake Free Air Gravity Anomaly Map

Figure 1 shows that the gravity field contains broad regional changes as well as smaller local features. These changes are important because gravity matching depends on being distinctive enough to separate one candidate location from another.

Local Gravity Variability

A local gravity variability map was calculated using the standard deviation in a 21 x 21 cell moving window. At a 500 m grid spacing this window represents an area approximately 10.5 km across. The purpose was to identify regions where the gravity field changes more strongly. The calculated local standard deviation ranged from approximately 0.375 mGal to 8.607 mGal, with a mean of approximately 2.538 mGal.

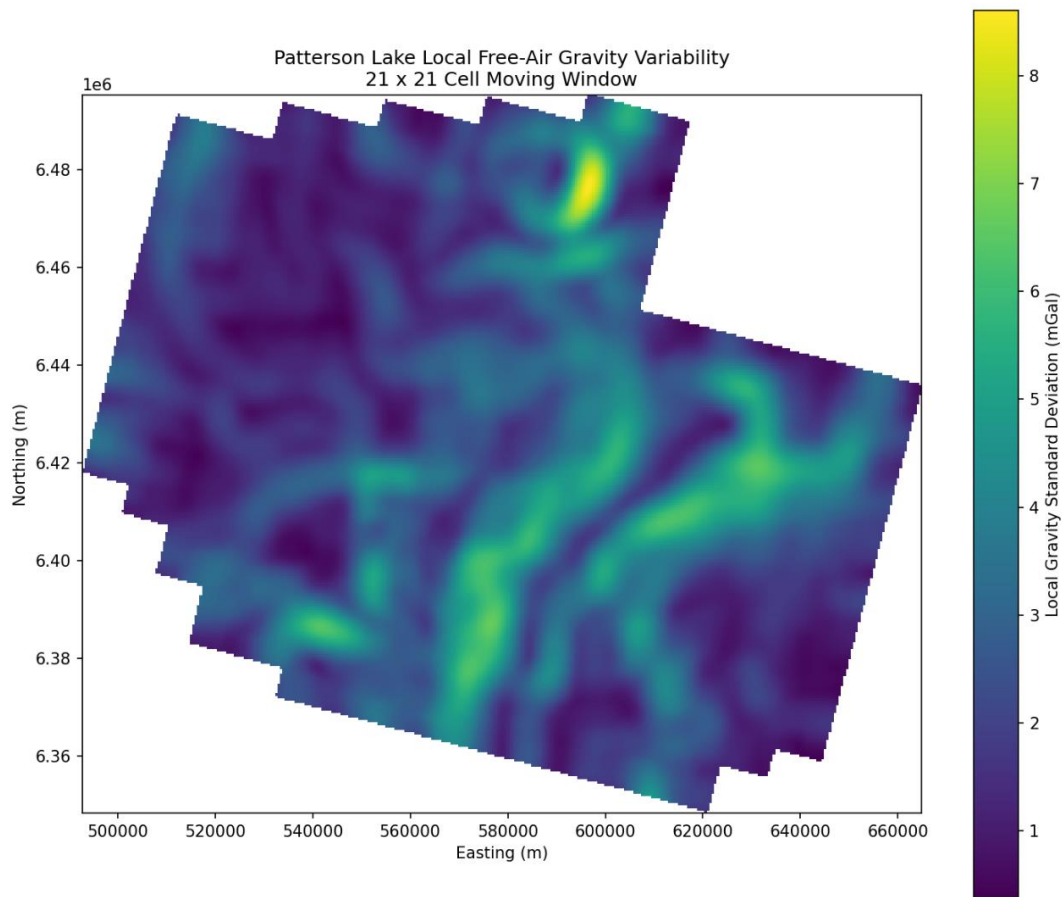


Figure 2: Patterson Lake Local Free Air Gravity Variability Using a 21 x 21 Cell Moving Window

First Vertical Derivative Gravity Layer

The First Vertical Derivative layer was added as a second channel because it emphasizes shorter wavelength changes in the gravity field. Figure 3 shows that this layer contains stronger local positive and negative features than the smoother Free Air anomaly map. This additional structure may make nearby locations easier to distinguish when the Free Air values alone are similar.

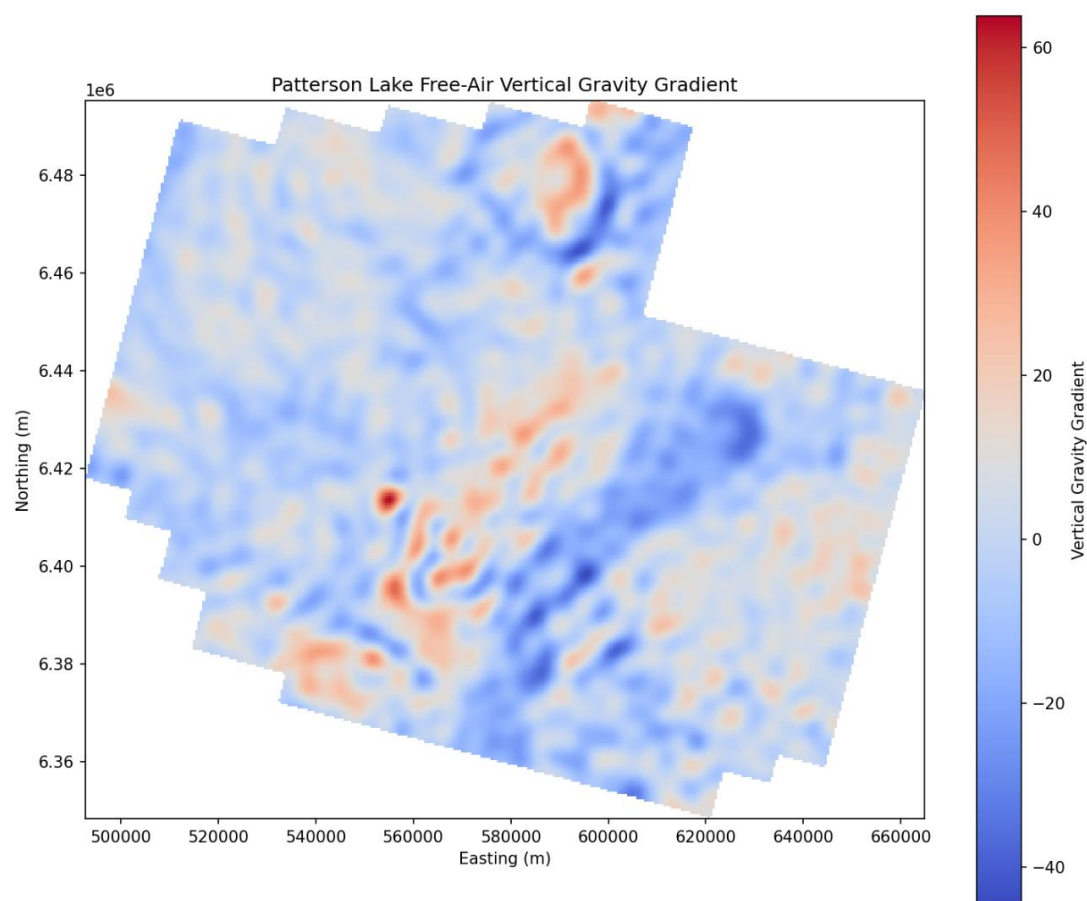


Figure 3: Patterson Lake First Vertical Derivative of the Free Air Gravity Anomaly

Methodology

The simulation was developed in stages so that each navigation component could be tested separately before combining the methods. The main workflow was: load and preprocess the gravity grids, generate a true aircraft trajectory, simulate gravity observations, generate an INS only path, apply TERCOM corrections, add the First Vertical Derivative map, implement OMSL-ICCP, and finally plan A* routes and apply the same OMSL-ICCP matching method to each route.

Simulation Configuration

Table 2: Main simulation configuration used for the final TERCOM comparison

Parameter	Value
Random seed	0
Aircraft speed	60 m/s
Gravity observation spacing	500 m
Free Air gravity white noise	0.5 mGal
Gravity bias standard deviation	0.5 mGal
Gravity bias correlation time	1800 s
First Vertical Derivative noise	3 E, treated as 0.3 mGal/km under the project unit assumption
Gyroscope bias	0.10 deg/hr
Accelerometer bias	50 micro-g
Initial position error	100 m
Initial velocity error	0.1 m/s
TERCOM matching window	30 km
TERCOM fix interval	5 km
TERCOM search radius	5 km
TERCOM search step	250 m

The final TERCOM simulation uses a gyroscope bias of 0.10 deg/hr. This value was increased from the earlier 0.05 deg/hr development setting to make the simulated INS drift more visible and to test the matching methods under a larger drift. OMSL validation and the final A* route experiments were completed earlier using the frozen 0.05 deg/hr development configuration. For this reason, the A* and OMSL results should be compared within their own experiment and should not be directly combined with the final 0.10 deg/hr TERCOM values as if they came from one identical run.

True Aircraft Path and Gravity Observations

The true aircraft path was created from five control points and interpolated using a cubic spline. The aircraft moves at 60 m/s and the trajectory is evaluated at 1 Hz. The original simulated path is approximately 150.83 km long and takes approximately 41.88 minutes. This produces 2,514 one-second trajectory positions.

Gravity observations are sampled every 500 m, producing 302 observations along the original flight path. At 60 m/s, the time between gravity observations is approximately 8.33 seconds. Free Air gravity values are interpolated from the stored grid at the true observation locations. White noise and a slowly varying bias are then added to represent a simplified gravimeter measurement model.

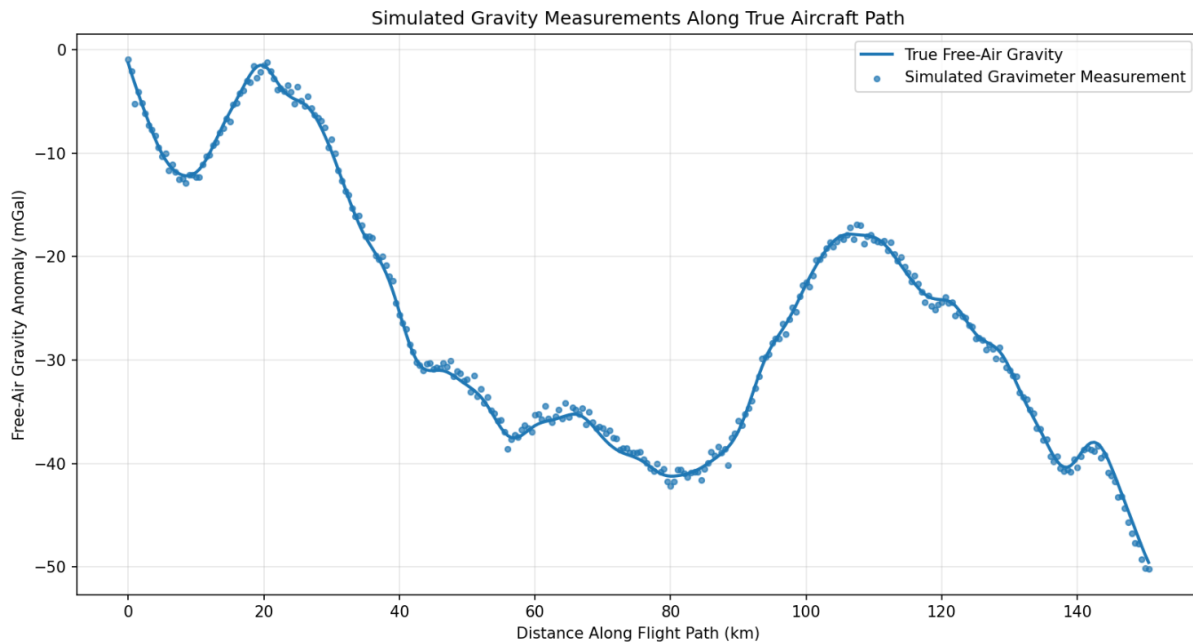


Figure 4: Simulated Free Air Gravity Measurements Along the True Aircraft Path

INS Error Simulation

The INS only path is generated using a simplified closed-form Schuler error model. The model is used to create a physically motivated, gradually growing horizontal drift that can be corrected by the gravity matching algorithms. The Schuler frequency is calculated from standard gravity and Earth radius.

Accelerometer bias and gyroscope bias are then converted into position-error terms with Schuler behaviour. Initial position and velocity errors are also included. The final 0.10 deg/hr configuration produces a final INS only horizontal position error of approximately 8.69 km on the original 150.83 km path. This larger error is useful for testing whether the map matcher can continue to recover the route as the INS estimate moves farther from the truth.

Free Air TERCOM Map Matching

The first map matching method is a TERCOM-style sequence matcher using only the Free Air gravity anomaly. At each correction point, the algorithm takes the most recent 30 km of gravity observations and searches candidate horizontal shifts around the current INS position. Because observations are spaced every 500 m, the matching window contains approximately 60 measurements.

Candidate shifts are evaluated within a 5 km search radius in both Easting and Northing using a 250 m step. This gives 41 candidate offsets in each direction, or 1,681 candidate shift combinations at each fix. For

every candidate, the measured gravity sequence is compared with the map gravity sequence. The candidate with the lowest mismatch is selected as the TERCOM correction. Corrections are generated every 5 km after enough data is available to fill the matching window.

The discrete corrections are then used to construct a continuous gravity corrected trajectory. This allows horizontal error to be evaluated continuously rather than only at the individual TERCOM fix locations.

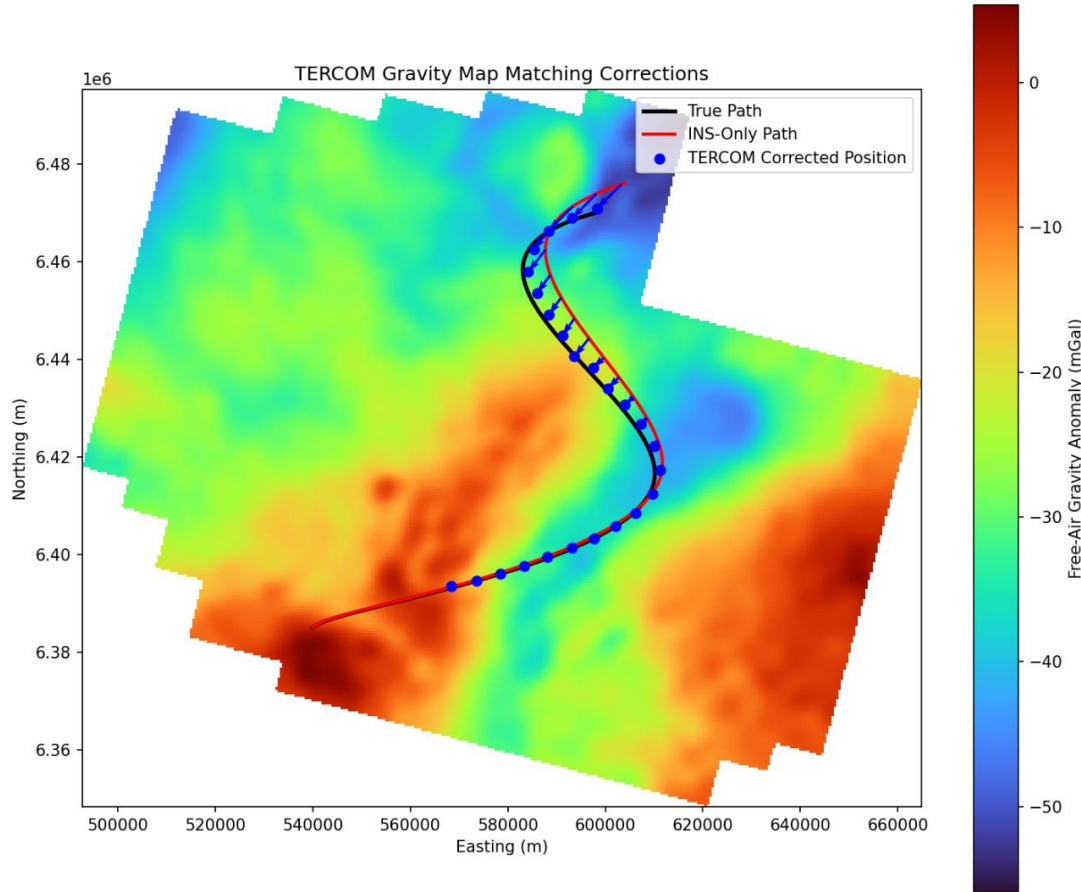


Figure 5: Free Air TERCOM Gravity Map Matching Corrections

Free Air & First Vertical Derivative TERCOM

The second TERCOM method adds the First Vertical Derivative as another measurement. The Free Air anomaly and derivative sequences are both evaluated at the same candidate positions. The method combines the two map mismatch costs after scaling so that the second layer contributes additional spatial information without replacing the Free Air anomaly. The goal is to test whether the sharper local structure in the derivative map helps reduce ambiguity between nearby candidate positions.

The same 30 km window, 5 km fix interval, 5 km search radius, and 250 m search step are used so that the comparison with the Free Air-only method remains consistent.

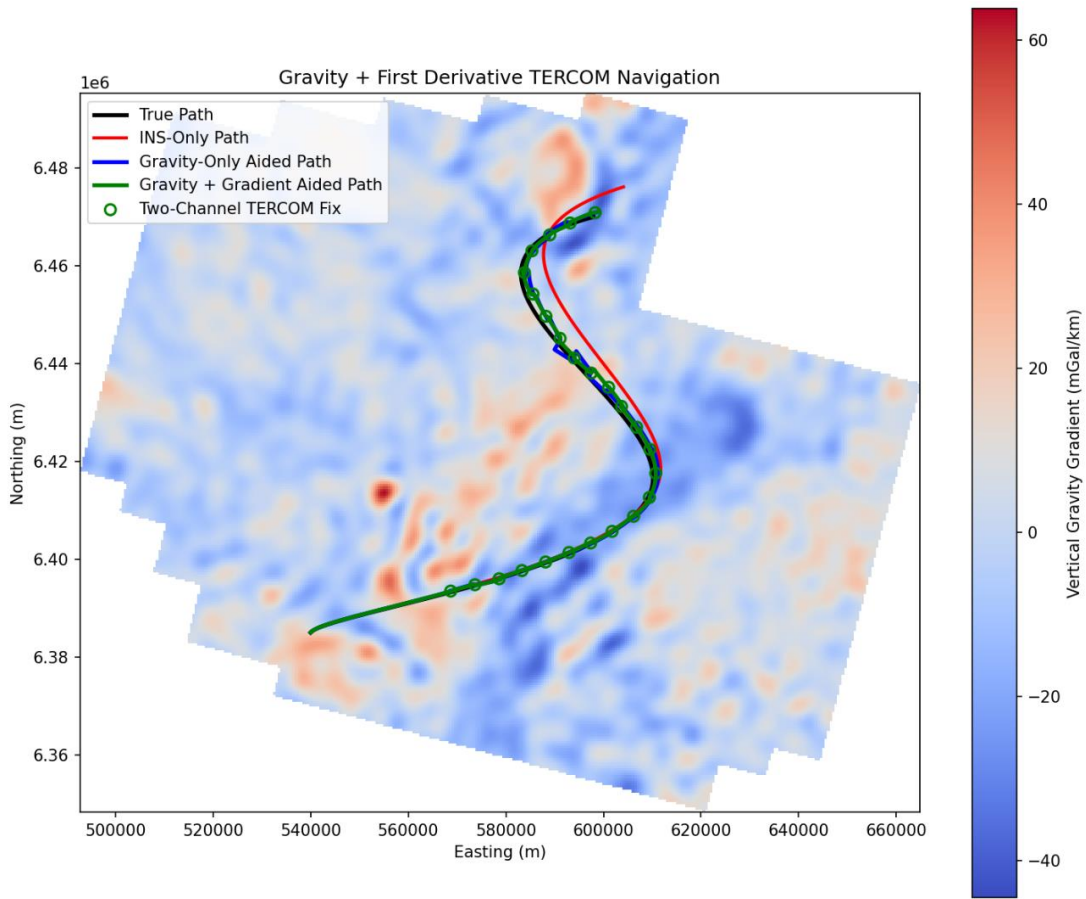


Figure 6: Free Air & First Vertical Derivative TERCOM Navigation

Sensitivity Testing

A sensitivity analysis was carried out during development to check how strongly the TERCOM result depends on the selected simulation settings. The tested factors included gravity noise, gravity bias, search radius, correction interval, and other matching parameters. This stage was used to identify whether a method only worked for one narrowly tuned setting. The sensitivity work also showed why search radius is a practical limitation. If INS drift becomes larger than the candidate search window, the correct solution may no longer be available to the matcher even when the gravity sequence itself is informative.

OMSL-ICCP Implementation

OMSL-ICCP was implemented after the TERCOM baseline. ICCP attempts to align the geometric shape of a measured gravity sequence with the closest gravity contour points in the reference map. The real-time form operates on a moving sequence of observations. OMSL changes the sequence length instead of keeping one fixed value for the full route [10].

The implemented search follows the Golden Section Search structure described by Liu et al. The search interval is 6 to 20 observations. OMSL starts after the tenth gravity measurement. Two interior sequence lengths are evaluated using the 0.382 and 0.618 golden ratios and are rounded to integer lengths. The endpoint-to-contour objective is used to choose the sequence length. A Hausdorff-distance calculation controls the closest-contour search region, and a GFA refinement is then applied to the ICCP result.

Several validation scripts were used to test rotation handling, sequence-length behaviour, search radius, gradient variants, capture behaviour, and the effect of initialization. These tests were important because the

paper describes the method but does not provide the exact source implementation. The validation also showed that a small contour distance objective does not always mean a small geographic position error.

A* Route Planning With OMSL-ICCP

The final algorithm development stage worked on finding a reasonably short route that passes through gravity regions expected to produce lower navigation uncertainty, then run OMSL-ICCP on that route.

All A* routes use the same start and goal locations, approximately (540,000 m E, 6,385,000 m N) and (598,000 m E, 6,470,000 m N). The search uses neighboring grid cells and traces the final route using parent nodes. Route A is the shortest-path baseline. Routes B, C, and D change the path cost by adding different forms of gravity information.

Route A: Shortest A*

Route A uses standard A* and primarily minimizes route distance. It provides the baseline needed to determine whether the extra distance introduced by gravity-aware planning actually improves OMSL-ICCP performance.

Route B: PCRB Gravity-Aware A*

Route B adapts the PCRB gravity-aware planning idea from Wang and Cai [12]. Gravity information and predicted navigation uncertainty are added to the route cost so that the path is encouraged to pass through locations expected to provide stronger positioning information.

Route C: OMSL-Aware A*

Route C uses local gravity distinctiveness. The route cost favours areas where the local gravity field changes more strongly and may therefore be easier to identify during OMSL-ICCP matching.

Route D: Sequence-Aware A*

Route D evaluates gravity distinctiveness along a sequence of locations rather than using only a single cell measure. This was developed to better match the way OMSL-ICCP uses a sequence of gravity observations during the matching process.

Evaluation Metrics

Navigation performance is evaluated using horizontal position error. The main summary metric is Root Mean Square Error (RMSE). Maximum position error and final position error are also reported. For the A* tests, total route distance is included because a navigation-aware route is only useful if the improvement is not achieved by an unreasonable increase in travel distance.

Results

INS Only and Free Air TERCOM Results

With the final gyroscope bias set to 0.10 deg/hr, the INS only trajectory ends approximately 8.69 km from the true position. After the first TERCOM correction becomes available, the INS only RMSE is approximately 4.311 km. Free Air TERCOM reduces the gravity-aided RMSE to approximately 1.039 km. The maximum gravity aided error is approximately 2.001 km and the final gravity aided error is approximately 1.620 km. This is a 75.90% reduction in RMSE compared with the INS-only solution over the same evaluation period.

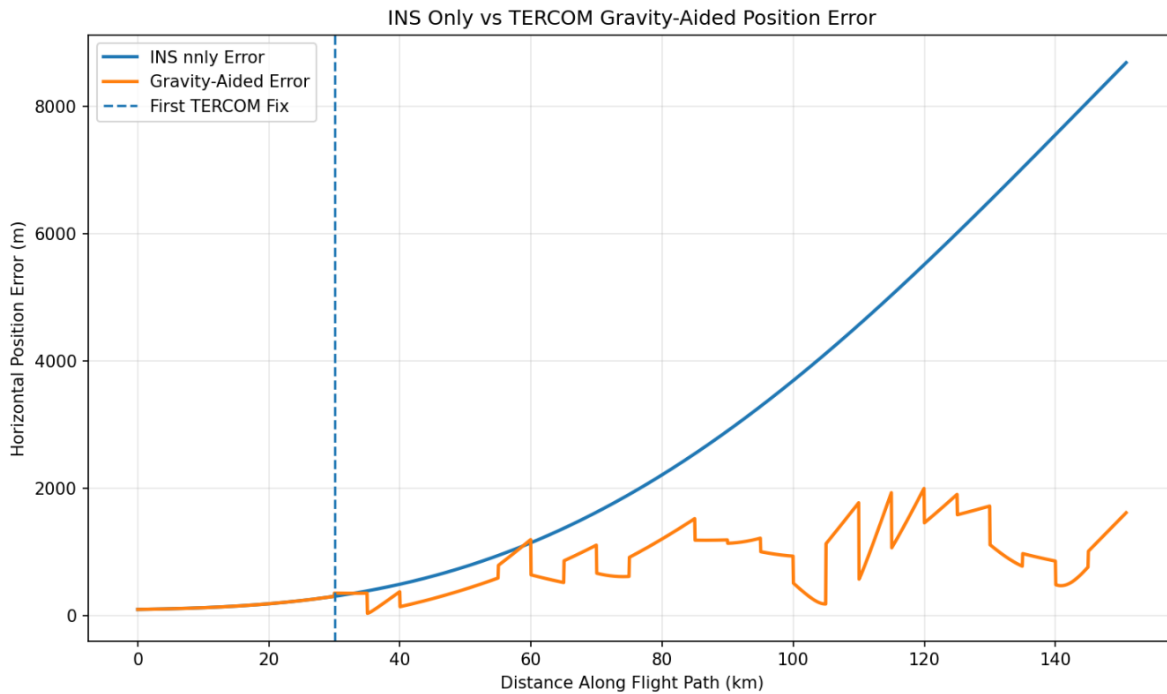


Figure 7: Horizontal Position Error for INS-Only and Free Air TERCOM

Free Air & First Vertical Derivative TERCOM Results

The two-channel method produces an RMSE of approximately 1.008 km after the first correction, with a maximum error of approximately 1.832 km and a final error of approximately 1.620 km. Compared with the same IN -only RMSE of 4.311 km, this is a 76.61% RMSE reduction.

A direct common-interval comparison from 45 km to the end of the route provides the clearest channel comparison. Over this interval, the INS only RMSE is 4.603 km, Free Air TERCOM is 1.105 km, and Free Air & First Vertical Derivative TERCOM is 1.071 km. The map solution therefore reduces the INS RMSE by approximately 76.7% and provides an additional improvement of approximately 3.1% compared with Free Air alone. The improvement from the derivative layer is real in this simulation but is much smaller than the improvement achieved by introducing gravity matching itself.

Table 3: Final TERCOM navigation results using the 0.10 deg/hr INS configuration

Method	RMSE after first fix (km)	Maximum error (km)	Final error (km)
INS only	4.311	8.691	8.691

Free Air TERCOM	1.039	2.001	1.620
Free Air & First Vertical Derivative TERCOM	1.008	1.832	1.620

Table 4: Common interval comparison from 45 km to the end of the flight path

Method	RMSE (km)	Change from INS only
INS only	4.603	Baseline
Free Air TERCOM	1.105	Approximately 76.0% lower
Free Air & First Vertical Derivative TERCOM	1.071	Approximately 76.7% lower

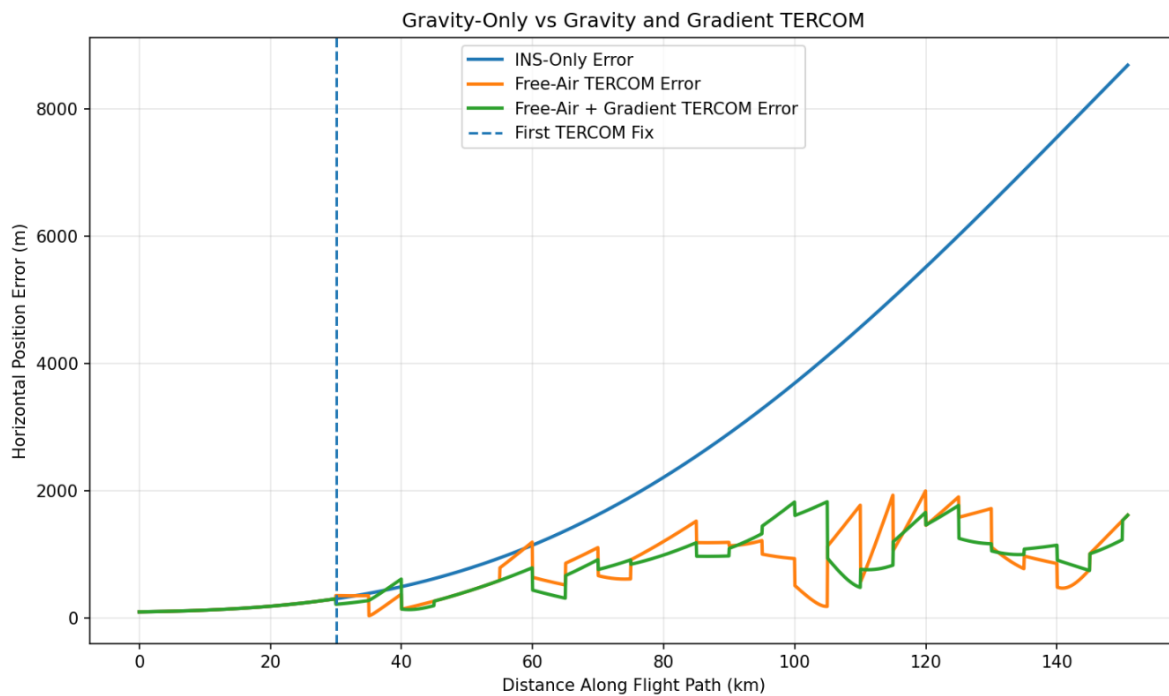


Figure 8: Position Error Comparison for Free Air TERCOM and Free Air & First Vertical Derivative TERCOM

TERCOM Search Radius Limitation

The larger final INS drift exposes an important limitation in the fixed TERCOM search window. Near the end of the route, the actual correction required is approximately -6.09 km in Easting and -6.09 km in Northing. The matcher is limited to a maximum candidate correction of -5 km in each direction. This means the true correction is outside the candidate set and the selected correction saturates at the search boundary. This result shows that increasing INS drift cannot be solved only by improving the gravity cost. A practical implementation also needs an adaptive search radius, a better predicted uncertainty region, or more frequent corrections.

OMSL-ICCP Validation Results

OMSL-ICCP was first tested using the raw simulated INS trajectory and noisy, biased gravity observations under the earlier 0.05 deg/hr development configuration. In this case the full OMSL-ICCP RMSE was approximately 1.754 km and the final position error was approximately 5.057 km. Although the method could produce good matches at some points, the sequence occasionally locked onto a geographically incorrect gravity contour. This caused the error to grow near the end of the route.

A second validation used the Free Air & First Vertical Derivative TERCOM trajectory as the initialization for OMSL-ICCP. With the same noisy gravity observations, the OMSL-ICCP RMSE improved to approximately 0.929 km and the final error improved to approximately 0.621 km. This large improvement shows that OMSL-ICCP is strongly affected by the quality of its initial path.

Table 5: OMSL-ICCP validation results under the frozen 0.05 deg/hr development configuration

OMSL initialization	OMSL RMSE (km)	Maximum error (km)	Final error (km)
Raw INS trajectory	1.754	5.170	5.057
Free Air & derivative TERCOM trajectory	0.929	2.933	0.621

A* Route Planning Results

The A* route experiments were completed using the frozen 0.05 deg/hr development configuration. Route A is the shortest path and is used as the baseline. Routes B, C, and D are slightly longer because they attempt to pass through gravity regions selected by their navigation-related cost functions.

Table 6: A route planning and OMSL-ICCP results*

Route	Planning method	Distance (km)	OMSL RMSE (km)	Maximum error (km)	Final error (km)
A	Shortest A*	109.024	1.307	3.274	2.185
B	PCRB Gravity-Aware A*	113.167	1.528	4.953	1.110
C	OMSL-Aware A*	112.752	1.701	6.188	1.059
D	Sequence-Aware A*	113.167	1.493	4.953	1.299

OMSL-ICCP Algorithms with Various Variations of A* Algorithm Animation

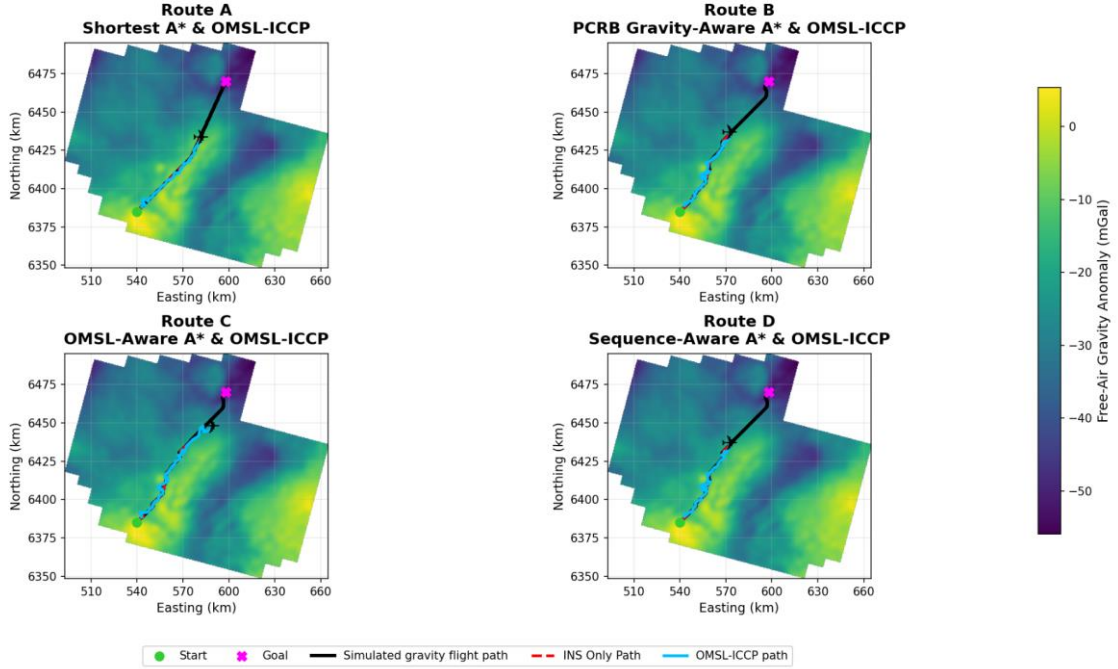


Figure 9: OMSL-ICCP Navigation for Routes A, B, C, and D

Route A produced the lowest overall OMSL-ICCP RMSE at approximately 1.307 km. Route D produced the best RMSE among the gravity-aware Routes B, C, and D at approximately 1.493 km. Route C produced the lowest final position error at approximately 1.059 km, even though it had the largest overall RMSE and maximum error. These results show that the selected route changes OMSL-ICCP performance, but none of the gravity-aware routes consistently outperformed the shortest baseline route.

Route B and Route D were approximately 3.80% longer than Route A, while Route C was approximately 3.42% longer. The extra distance was therefore relatively small. However, the additional travel did not produce a consistent RMSE benefit. This indicates that a planning metric based only on local gravity information or predicted uncertainty is not enough to guarantee a better sequence match.

Analysis

Effectiveness of Gravity Map Matching

The strongest result from the simulation is the reduction in INS error produced by TERCOM gravity matching. With the final 0.10 deg/hr INS configuration, both gravity-aided methods reduce RMSE by more than 75% compared with the INS-only solution. This shows that even a relatively simple sequence-matching method can control a large part of the simulated drift when the correct region remains inside the search window.

The position-error plots also show that the gravity corrected path does not continuously converge to zero error. Instead, error decreases after successful corrections and then grows again between fixes. This behaviour is expected because the underlying INS continues to drift. The gravity solution acts as a periodic external correction rather than replacing the INS propagation.

Value of the First Vertical Derivative

Adding the First Vertical Derivative improves the TERCOM result, but the effect is modest compared with the improvement from gravity matching itself. On the common interval, RMSE changes from 1.105 km for Free Air alone to 1.071 km for the two-channel method. The main benefit is therefore additional local discrimination rather than a completely different navigation solution.

The derivative layer can also introduce sensitivity because it contains sharper spatial variation and depends on the physical interpretation and noise model of the gradient data. Since the source unit was not explicitly confirmed in the metadata used by the code, the current gradient noise scaling remains a project assumption. A future physical implementation should verify the source unit and use a sensor-specific gradient noise model.

Implementation of A* & OMSL-ICCP Results

The A* results do not support the simple conclusion that routing through high-gradient or high-distinctiveness gravity regions automatically improves OMSL-ICCP. Route A, which is primarily the shortest route, achieved the best overall RMSE. Route D was the strongest gravity-aware route, and Route C ended closest to the true goal, but both still had larger overall RMSE than Route A.

This could be because planning routes and the OMSL matching objective are not perfectly aligned. Local gradient, PCRB information, or sequence distinctiveness can identify a region that appears informative, but these metrics do not directly measure whether the entire measured gravity sequence will be geographically unique. A route can therefore pass through strong gravity features and still encounter a repeated contour pattern that causes a false match.

This is still a useful negative result. It shows that route planning for gravity navigation should include a measure of global or sequence-level uniqueness, not only local variability. A future route cost could explicitly penalize gravity sequences that have similar matches elsewhere in the map.

Limitations

The main limitations of this research are:

- **The aircraft and INS are simulated.** The INS model creates realistic position drift, but it does not include a complete INS system with full attitude, velocity, and sensor error calculations.

- **The gravity measurements are simulated.** They are created from the same gravity map used for matching, with added noise and bias. Real airborne gravity measurements would contain additional errors caused by aircraft motion, changing height, sensor behaviour, map errors, and other environmental effects.
- **TERCOM uses a fixed 5 km search radius.** Near the end of the simulation, the INS drift becomes larger than this search area. This means that the correct position may fall outside the area being searched.
- **The gravity maps have a grid spacing of approximately 500 m.** This means that smaller gravity changes between grid cells cannot be represented. A higher resolution gravity map may provide more detailed information for map matching.
- **Only the Patterson Lake gravity survey area was tested.** The algorithms may perform differently in other locations. Areas with very smooth gravity fields may be more difficult to use for gravity map matching.
- **Only a limited number of simulated routes were tested.** The results show how the methods performed for the selected routes, but they do not represent every possible aircraft path or direction of travel.
- **The stored gravity map is treated as the reference truth.** Errors in the gravity map itself were not added to the simulation. Real gravity maps can contain measurement, interpolation, and resolution errors.
- **The TERCOM settings remain fixed during the simulation.** The matching window, search radius, correction interval, and search step do not automatically change when the INS uncertainty becomes larger or smaller.
- **The OMSL-ICCP code is a paper-based implementation.** It was created using the method and parameters described by Liu et al. (2019), but the authors' original source code was not available. Therefore, the implementation may not be exactly the same as the original algorithm.
- **OMSL-ICCP can sometimes match to the wrong location.** Similar gravity patterns can exist in different areas of the gravity map. Because of this, a gravity sequence can appear to be a good match even when the estimated position is geographically incorrect.
- **The gravity-aware A* routes did not always give better navigation results.** Route A, which mainly followed the shortest path, still had the lowest overall OMSL-ICCP RMSE. This shows that travelling through areas with stronger gravity information does not automatically guarantee a better position match.
- **The simulation mainly evaluates horizontal position.** Full three-dimensional aircraft movement, changing altitude, aircraft attitude, and realistic flight dynamics are not fully included.
- **The methods were tested in the same general simulation environment in which they were developed.** Testing the algorithms using a different gravity survey area and real airborne gravity measurements would provide stronger validation of the results.

Conclusion

The research successfully developed a complete simulation for testing Gravity Aided Navigation in a GNSS-denied environment. Real Patterson Lake airborne gravity data was used to create a known aircraft route, simulated gravity observations, a drifting INS solution, gravity map matching corrections, OMSL-ICCP matching, and gravity-aware A* route planning.

The Free Air TERCOM method showed that gravity map matching can greatly reduce simulated INS drift. Under the final 0.10 deg/hr INS configuration, the INS only RMSE after the first correction point was approximately 4.311 km, while Free Air TERCOM reduced the RMSE to approximately 1.039 km. Adding the First Vertical Derivative reduced it further to approximately 1.008 km. On the common comparison interval,

the two-map solution reduced RMSE by approximately 76.7% compared with INS only and by approximately 3.1% compared with Free Air TERCOM alone.

The OMSL-ICCP work showed that sequence-based gravity matching can provide useful position corrections, but the method is sensitive to initialization and false contour matches. Using the TERCOM corrected trajectory as OMSL initialization reduced the OMSL RMSE from approximately 1.754 km to 0.929 km and reduced the final error from approximately 5.057 km to 0.621 km. This showed that OMSL can be stabilized by a better initial trajectory, but it did not replace the need for map matching.

The A* route planning showed that gravity information can change the selected route and can change OMSL-ICCP performance. However, the gravity-aware routes did not consistently outperform the shortest Route A. Route D gave the lowest RMSE among the gravity-aware routes, while Route C gave the lowest final error. The result indicates that a good gravity navigation route cannot be selected only from local gravity strength or variability but also be mindful to whether the sequence is geographically unique and resistant to false matching.

Overall, the project supports gravity data as a useful independent aid for INS during GNSS denied environments and identifies two important directions for improvement: by changing the gravity search area based on how uncertain the position is and by choosing routes with more unique gravity patterns that are easier to match.

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